

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
library(tidyverse)
library(lubridate)
library(here)

getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here() #check working directory
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Neonics <- read.csv(  
  file = here('Data','Raw','ECOTOX_Neonicotinoids_Insects_raw.csv'),  
  stringsAsFactors = T  
) # upload dataset  
  
Litter <- read.csv(  
  file = here('Data','Raw','NEON_NIWO_Litter_massdata_2018-08_raw.csv'),  
  stringsAsFactors = T  
) # upload dataset
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: 1. we can understand how neonicotinoids affect non-target species and biodiversity, especially if its going to affect the pollinators since this has been widely used in agricultural fields.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: litter and woody debris play a crucial role in nutrient cycling. They also store carbon and provide habitat for many species, espacially insects.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: They set up traps in the forest to collect debris and litter. 1. Traps: elevated traps and ground traps capture debris and litter in different sizes 2. spatial layout: Traps are placed in forested "Tower plots". Each plot has 1-4 trap pairs (elevated + ground). 3. Elevated traps: every 2 weeks in deciduous sites (leaf-fall), every 1-2 months in evergreen sites. Ground traps: once per year. Subset of samples analyzed for chemistry/isotopes every 5 years.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics) #use the dim() function to see the dimension of the dataset
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect) #summary the specific column within the dataset
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##             12             102             360             11
##      Cell(s)      Development      Enzyme(s)      Feeding behavior
##             9             136             62             255
##      Genetics      Growth      Histology      Hormone(s)
##            82             38             5             1
##      Immunological      Intoxication      Morphology      Mortality
##            16             12             22             1493
##      Physiology      Population      Reproduction
##             7             1803             197
```

```
sort(table(Neonics$Effect), decreasing = T) #sort the column by decreasing order
```

```
##
##      Population      Mortality      Behavior      Feeding behavior
##            1803            1493            360            255
##      Reproduction      Development      Avoidance      Genetics
##            197            136            102            82
##      Enzyme(s)      Growth      Morphology      Immunological
##             62             38             22            16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##             12             12             11             9
##      Physiology      Histology      Hormone(s)
##             7             5             1
```

Answer: Population, Mortality, Behavior and Feeding behavior are the top 4 most common effects that are studied. population: shows how the number of the species change across time, the most direct way of seeing how Neonicotinoids is affecting the species. Mortality: shows the death rate, easy to see if Neonicotinoids is detrimental to the insects we are studying. Behaviors: some insects, especially pollinators rely on complex behavior to communicate, if Neonicotinoids will affect their behavior, it could disturb the way insects act. Feeding behavior: feeding is tied to growth, reproduction and many other aspect of the species. If the insects are not eating anymore, it could cause rapid death.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 6) # only showing the top 6,
```

```
##           Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee      Bumble Bee      (Other)
##           152           140           3196
```

```
#saving space in the knitted file
```

```
head(sort(table(Neonics$Species.Common.Name), decreasing = T)) #shows the top 6
```

```
##
##           Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##           152           140           113
```

```
#most commonly studied species
```

Answer: Honey Bee, Parasitic Wasp, Buff Tailed Bumble bee, and Carniolan Honey Bee are the top four most commonly studied species. All these species are pollinators which means they play a significant role in the agricultural ecosystem. It makes sense to study them in order to understand how Neonicotinoids is affecting the insects community.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

```
view(Neonics)
```

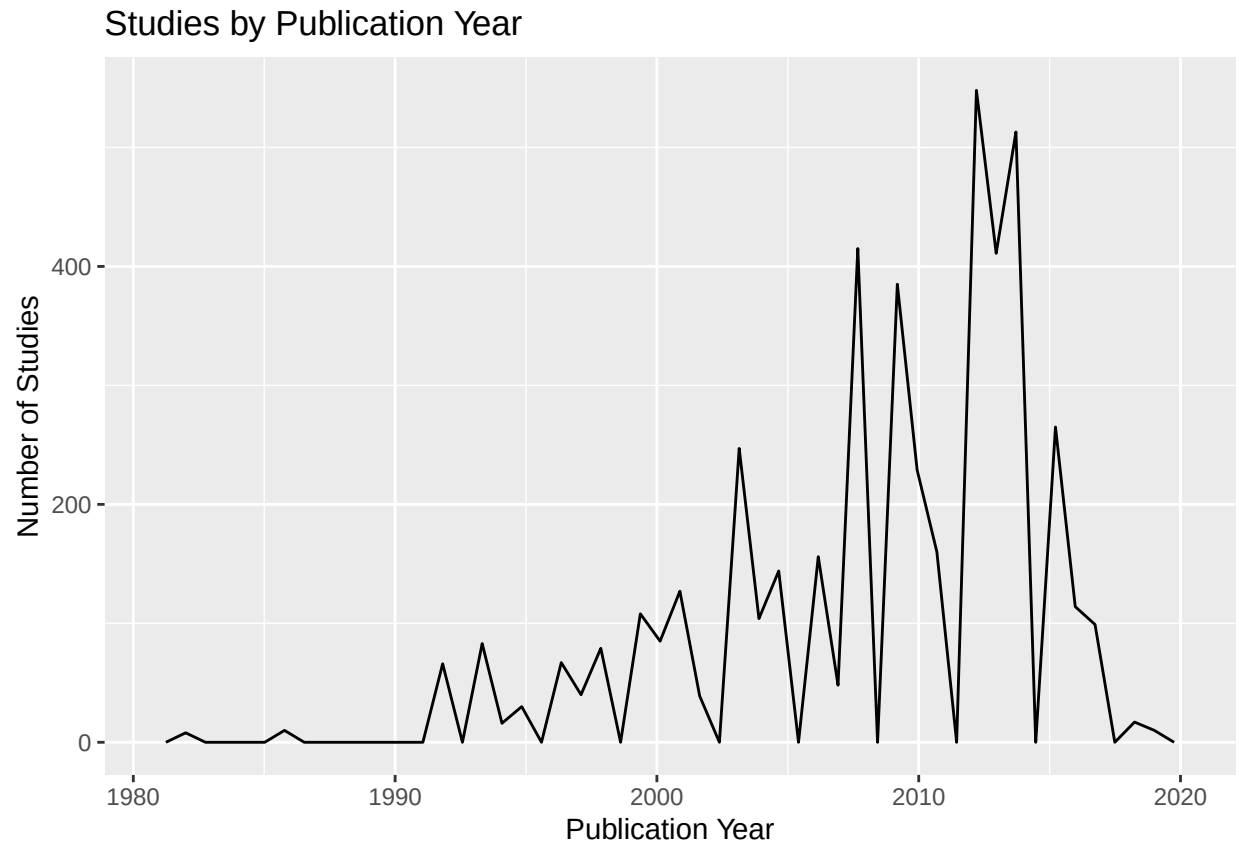
Answer: There are characters (NR) and symbols (/) within the column.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
library(ggplot2)

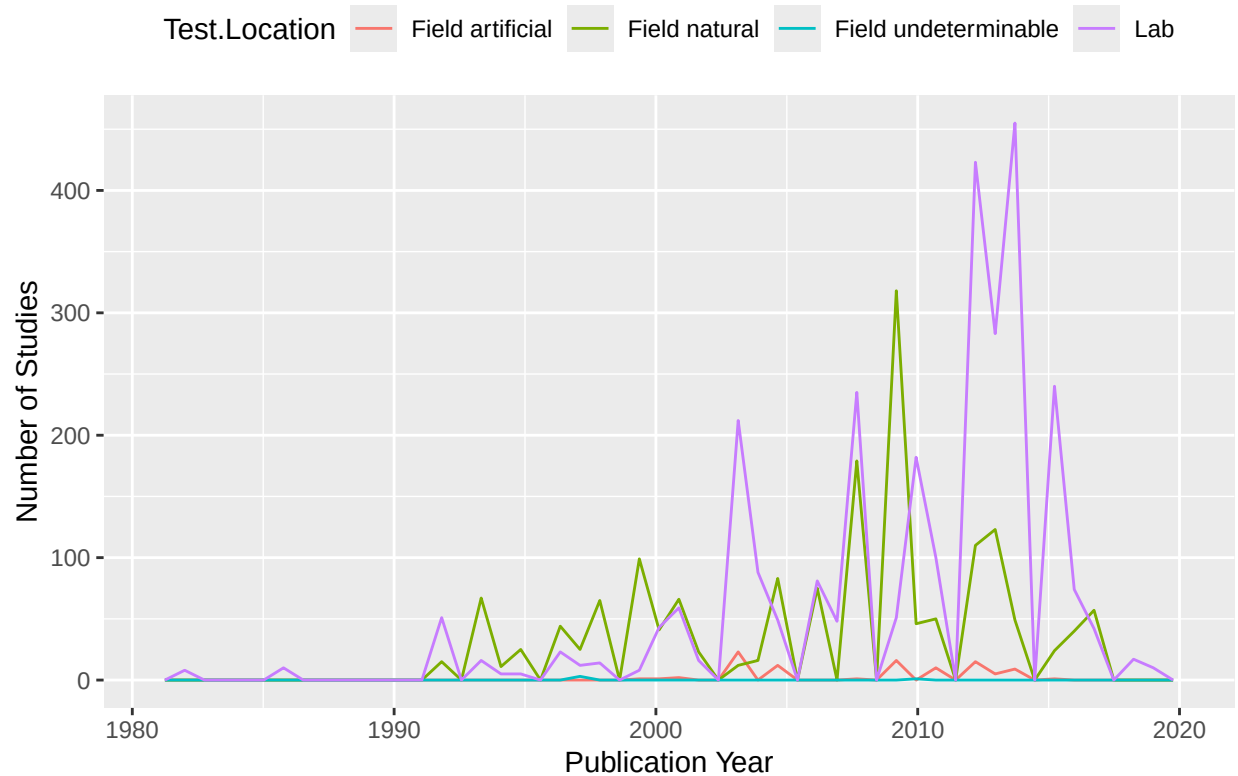
ggplot(Neonics) +
  geom_freqpoly(aes(x = Publication.Year), bins = 50) +
  scale_x_continuous(name = "Publication Year") +
  labs(y = "Number of Studies",
       title = "Studies by Publication Year")
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics) +
  geom_freqpoly(aes(x = Publication.Year, color = Test.Location), bins = 50) +
  scale_x_continuous(name = "Publication Year") +
  labs(y = "Number of Studies",
       title = "Studies by Publication Year by Test Location") +
  theme(legend.position = "top")
```

Studies by Publication Year by Test Location



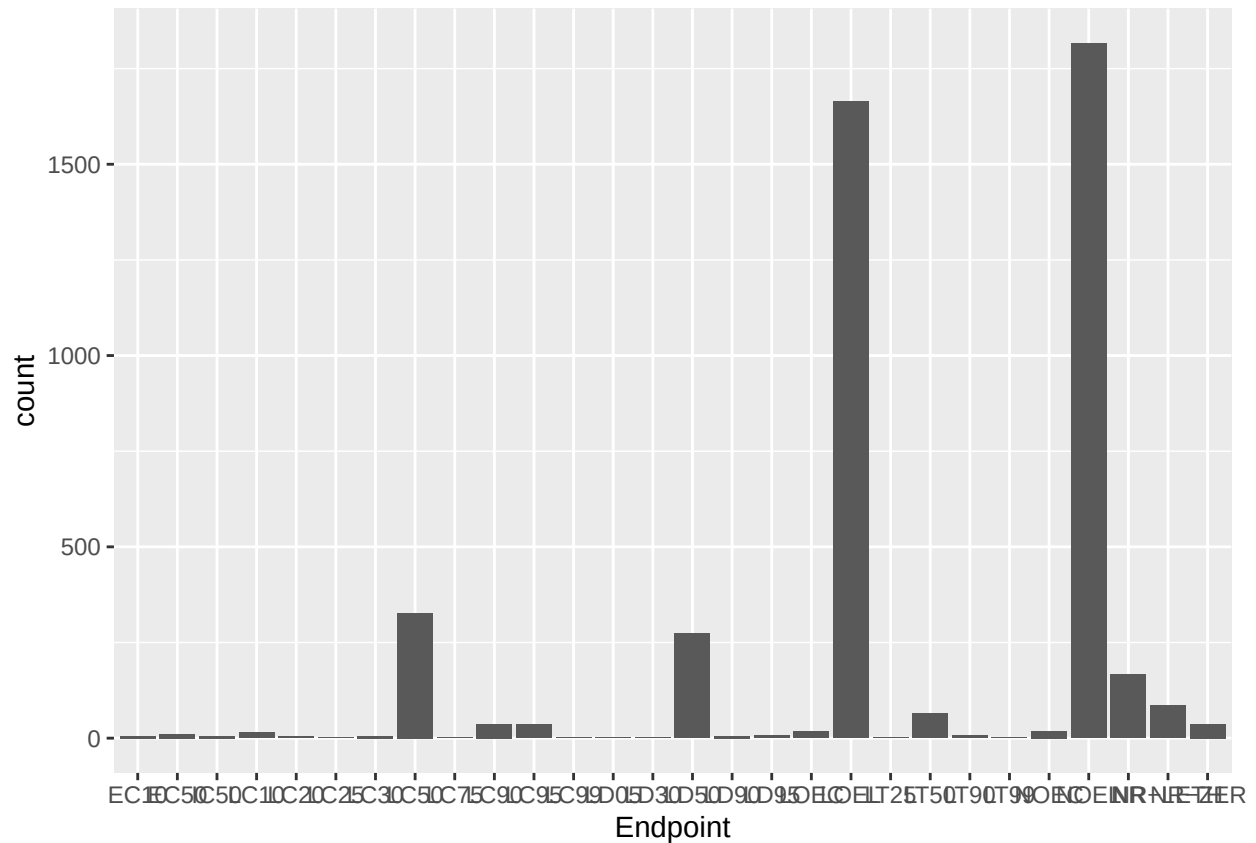
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test location is in the lab. They do differ over time. Between 1990 and 2010, both lab test and field natural test are common, after 2010, there is a great increase in the number of studies done in the lab.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(data = Neonics, aes(x = Endpoint)) +  
  geom_bar()
```



```
theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

```
## <theme> List of 1
## $ axis.text.x: <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight   : num NA
## ..@ fontwidth    : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : num 1
## ..@ vjust       : num 0.5
## ..@ angle       : num 90
## ..@ lineheight   : NULL
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi FALSE
## @ complete: logi FALSE
## @ validate: logi TRUE
```

Answer: MOR → Mortality Definition: death of organisms following exposure to a chemical.
GRO → Growth Definition: change in size, weight, or developmental progression of organisms.

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate) #check the class, it's in factor
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(Litter$collectDate, format = "%Y-%m-%d") #convert into date  
head(Litter$collectDate) #check the first few lines
```

```
## [1] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"  
## [6] "2018-08-02"
```

```
class(Litter$collectDate) #check the class again
```

```
## [1] "Date"
```

```
unique(Litter$collectDate[  
  format(Litter$collectDate, "%Y-%m") == "2018-08"  
) #find the date within 2018-08.
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

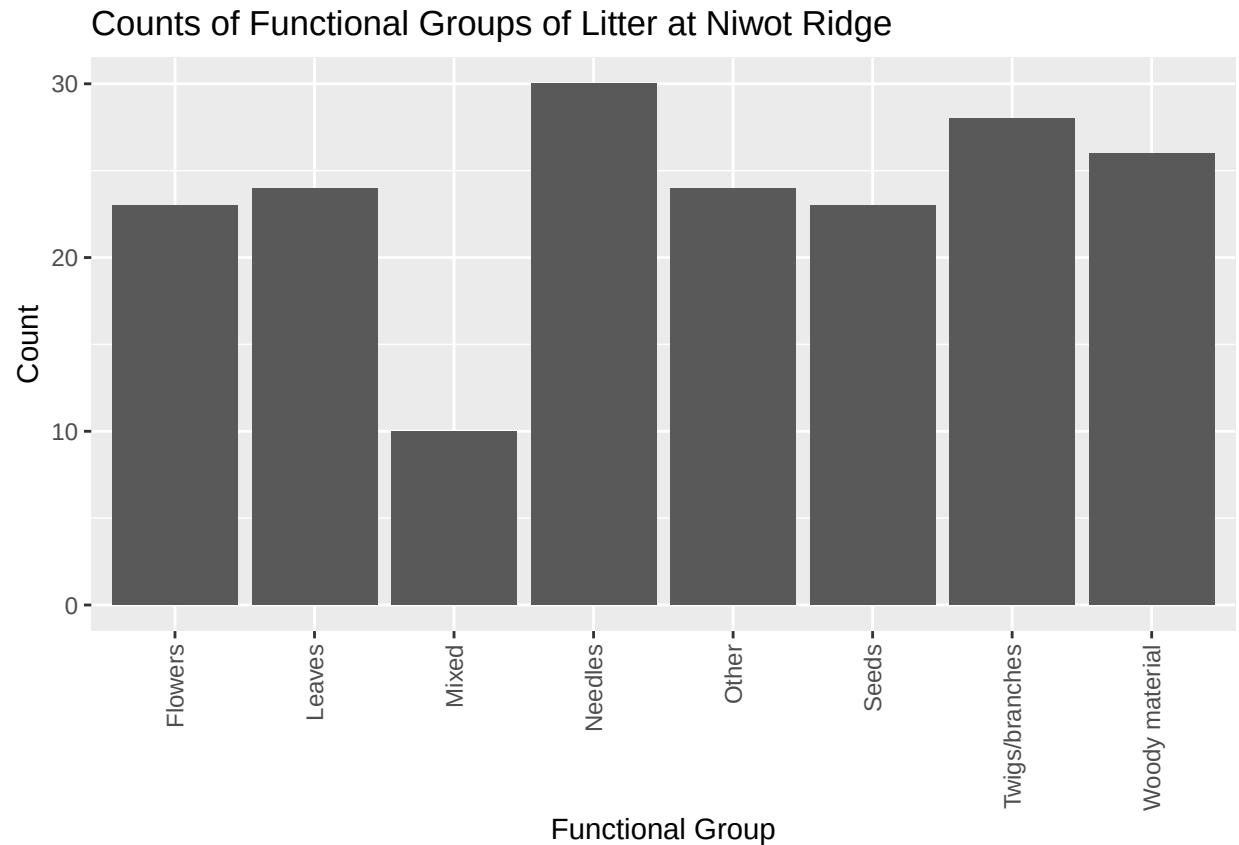
```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: `unique` function only gives you the set of distinct values only. Also, it does not tell you how many times each value occurs.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

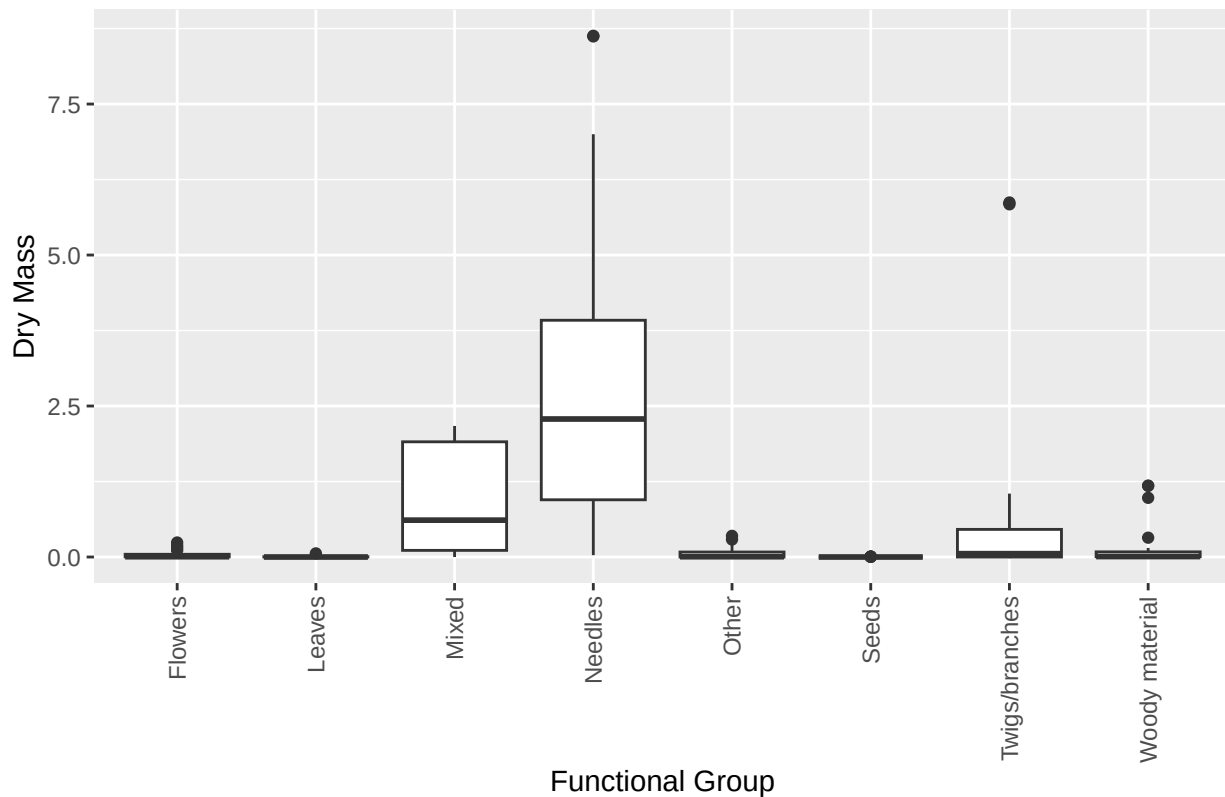
```
ggplot(data = Litter, aes(x = functionalGroup)) +  
  geom_bar() +  
  labs(title = "Counts of Functional Groups of Litter at Niwot Ridge",  
        x = "Functional Group",  
        y = "Count") +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
ggplot(data = Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_boxplot() +  
  labs(title = "Boxplot of Dry Mass by Functional Group",  
        x = "Functional Group",  
        y = "Dry Mass") +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

Boxplot of Dry Mass by Functional Group



```
ggplot(Litter) +
  geom_violin(aes(x = functionalGroup, y = dryMass),
    draw_quantiles = c(0.25, 0.5, 0.75)) +
  scale_y_log10() +
  labs(title = "Dry Mass by Functional Group (log scale)",
    x = "Functional Group", y = "Dry Mass (log10)") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

```
## Warning: The 'draw_quantiles' argument of 'geom_violin()' is deprecated as of ggplot2
## 4.0.0.
```

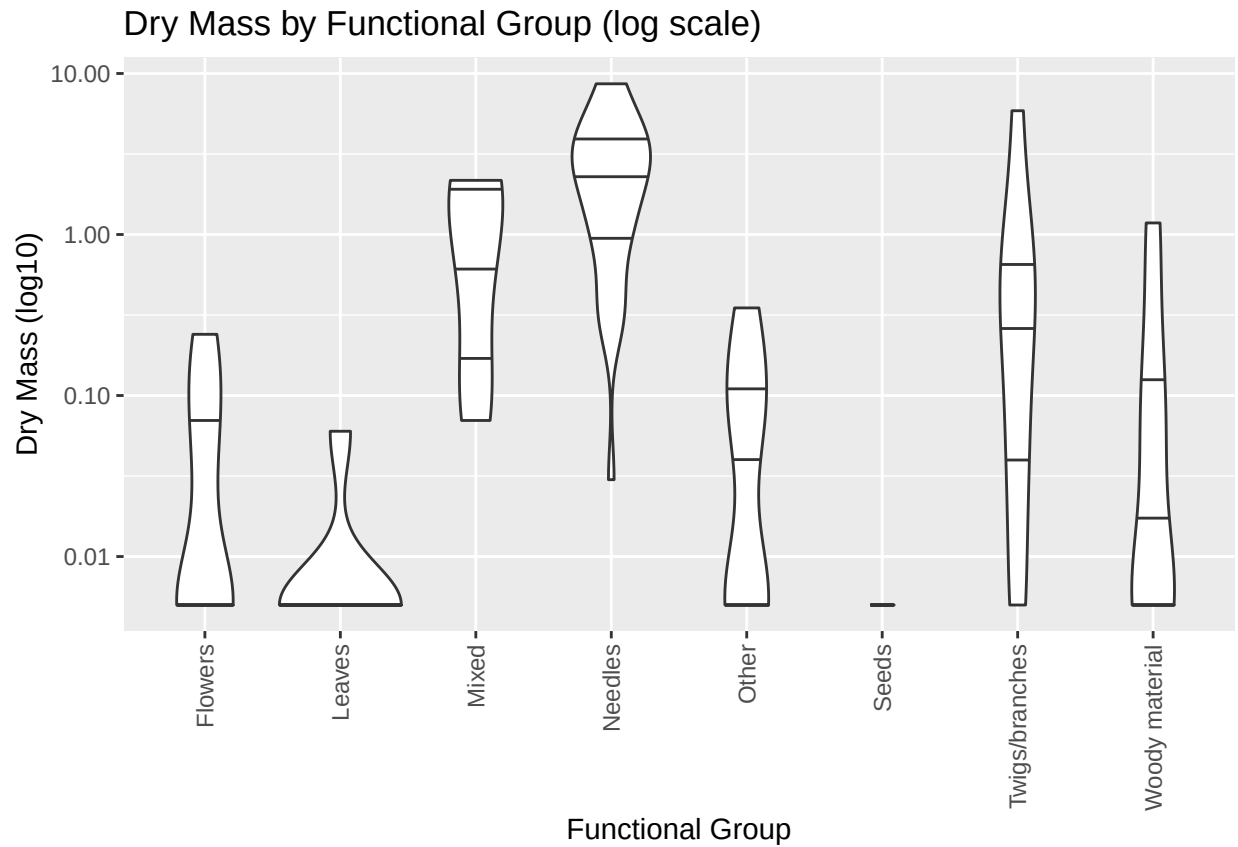
```
## i Please use the 'quantiles.linetype' argument instead.
```

```
## This warning is displayed once every 8 hours.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 61 rows containing non-finite outside the scale range
## ('stat_ydensity()').
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: Because boxplot is more simple and effective in terms of representation.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles tend to have the highest biomass at these site since it has the highest dry mass.